

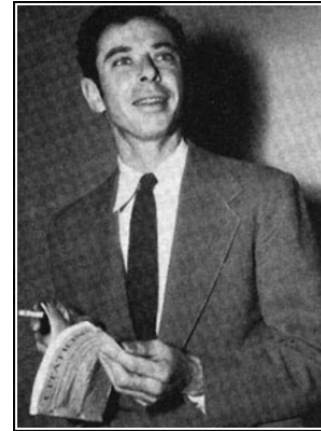
CS 59300

Application of Deep Learning Deep Learning Methodology

Instructor: Dr. Zesheng Chen

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Why I Teaching This Course?



The best way to learn is to teach

— Frank Oppenheimer —

AZ QUOTES

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Learning Methodology

- Learn from examples
 - Machine Learning: "learn from data"
- Compare with something familiar with
 - Deep learning: "transfer learning"
- Do it! Try it out!
 - "Trial and error"!
- Reading (books or papers)!
- Work on a project!

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Learning Methodology (Cont)

- Work with someone who is smarter than you!
- Reflection! Take notes!
 - What are mistakes to avoid in the future!
 - What are good practices to keep!
 - My own teaching experience
 - Take notes for each lecture

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Learning Methodology

- Two learning methods:
 - Learning by instruction
 - Learning by discovery
- Software development
 - Ability to find the answer quickly
 - Ability to learn how to find the answer quickly
 - Ability to ask good questions

Learningability is very important!!!

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Learning Methodology

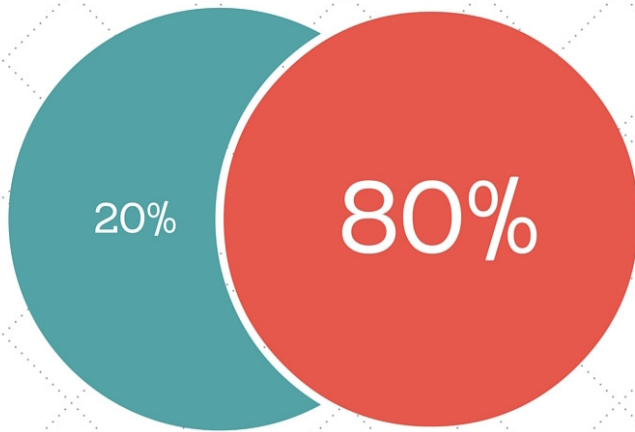
- Attend the class
- Join the discussion
- Review course materials in time
- Understand course materials and homework
- Work on homework problems
- Work on course project
- Ask questions
- Practice! Practice! Practice!

This course takes time!!!

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THE 80/20 RULE



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What are methods to effectively and efficiently **find** the answer to a question?

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Different Methods

- ChatGPT (or other AI tools)
- Google (or Bing or other search tools)
- Books
- People

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Prompt Engineering

- https://en.wikipedia.org/wiki/Prompt_engineering
- A nice free course:
- “ChatGPT Prompt Engineering for Developers”
- <https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>



Google™ Is Your Best Friend ...

- Look at:
 - http://www.google.com/advanced_search
 - <http://www.sitepoint.com/10-tips-for-conducting-a-more-effective-google-search/>

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AI Tools Policy

- AI tools
 - For example, ChatGPT, Colab AI Assistant, or Amazon Codewhisperer
- Good to use AI tools to **learn** course materials
 - Ask questions about concepts or technologies
 - Ask for more code examples
 - Ask for the error messages from running your code



AI Tools Policy

■ HW assignments

- **No AI tools** for code generation or answer questions, except required by the homework.
- You need to learn the **fundamental** knowledge.
- But it can ask AI tools or Google on the error messages or debugging the code.

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AI Tools Policy

■ Course project

- Feel free to use AI tools.
- If the code is generated by a AI tool, indicate this in the comment section of the program.
- **Suggestion**: “If you are sure that you can implement it **without** an AI tool, go ahead to use AI. Otherwise, please do **NOT** use AI.”
- Vibe coding can be “**harmful**” to your project.

Q & A